IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

<Name>
<Date>

Outline

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Executive Summary

- This project analyzes Falcon 9 first-stage landing outcomes using a complete end-to-end data science workflow. Launch data was collected from both the SpaceX REST API and Wikipedia, then cleaned, merged, and prepared for analysis. Exploratory data analysis and interactive dashboards revealed clear relationships between landing success and key factors such as launch site, payload mass, and booster version.

- Four classification models—Logistic Regression, SVM, KNN, and Decision Tree—were developed to predict landing success. All models achieved identical performance (0.83 accuracy) and produced the same confusion matrix, reflecting the small test set and consistent patterns in the data. With no performance differences to separate the models, Logistic Regression was selected as the final model based on its simplicity, interpretability, and strong generalization characteristics.

- The final model correctly identified all successful landings and provides a reliable baseline for future improvements. Overall, this project demonstrates a complete data science pipeline, from data acquisition and wrangling through visualization, modeling, and insight generation.

Introduction

SpaceX’s Falcon 9 rocket relies on reusable first-stage boosters to reduce launch costs. Predicting whether a booster will successfully land is therefore an important operational and economic question.

This project uses a complete data science workflow—data collection, cleaning, exploration, visualization, and machine learning—to analyze the factors that influence landing success and to build a predictive model.

Launch data was gathered from the SpaceX API and Wikipedia, prepared for analysis, and explored using SQL, visualizations, and interactive dashboards. Multiple classification models were then developed to predict landing outcomes, with Logistic Regression selected as the final model based on its simplicity and generalization

Section 1

Methodology

Methodology

Data Collection:

- Retrieved launch records from the SpaceX REST API
- Scraped supplemental mission details from Wikipedia

Data Wrangling:

- Cleaned, merged, and standardized datasets
- Engineered features for analysis and modeling

Exploratory Data Analysis (EDA):

- Used SQL queries to explore launch outcomes and site patterns
- Created visualizations to examine payload, booster version, and success rates

Interactive Visual Analytics:

- Built a Folium map to visualize launch sites and outcomes
- Developed a Plotly Dash dashboard for interactive exploration

Predictive Analysis:

- Trained Logistic Regression, SVM, KNN, and Decision Tree models
- Tuned and evaluated models using accuracy and confusion matrices
- Selected Logistic Regression based on simplicity and generalization

Data Collection

Data Sources:

- SpaceX REST API for launch records and mission details
- Wikipedia scraping for supplemental booster and landing information

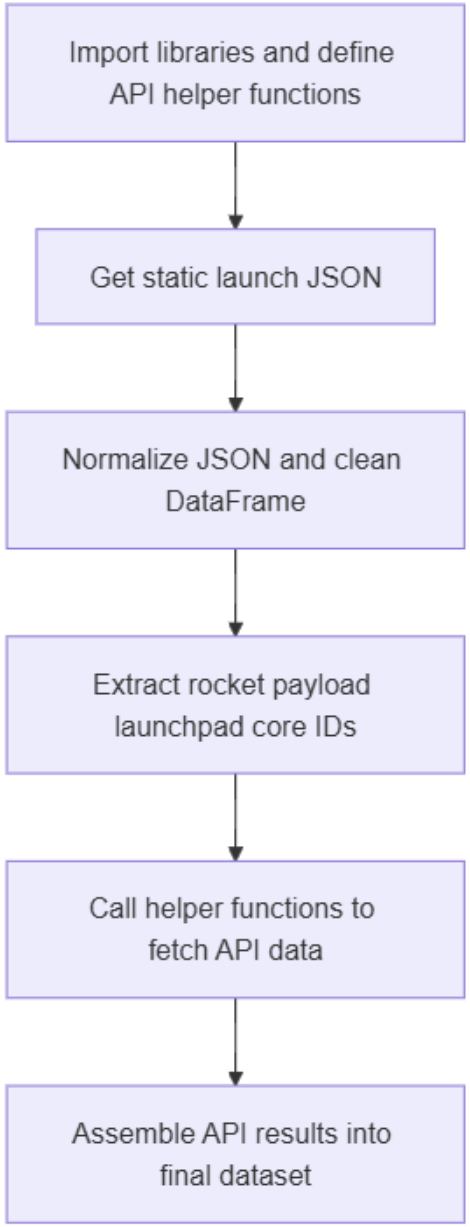
Collection Process:

- Queried the SpaceX API to retrieve structured JSON launch data
- Extracted and parsed HTML tables from Wikipedia using web scraping
- Combined both sources to create a complete dataset for analysis

Data Collection – SpaceX API

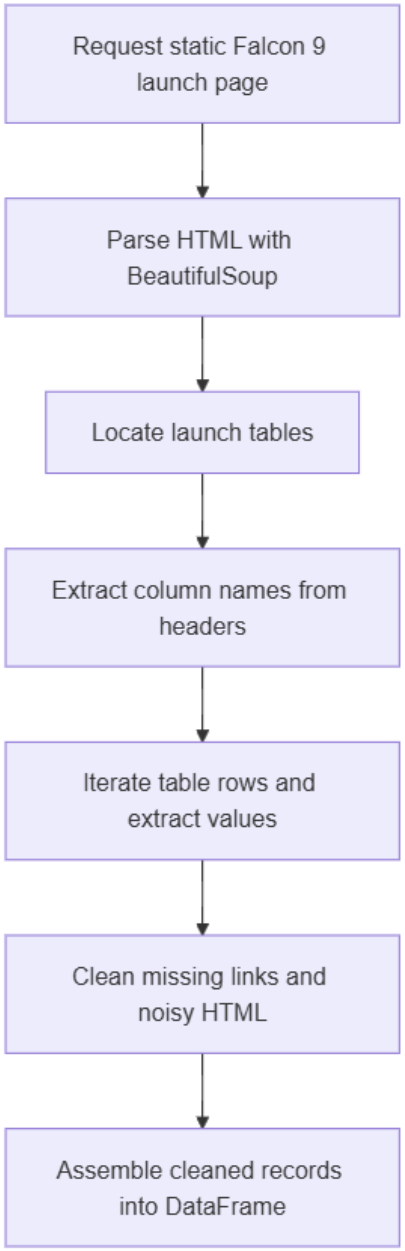
- Collected SpaceX launch data from a static JSON file and enriched it with REST API calls to rockets, payloads, launchpads, and cores.
- Queried each endpoint using helper functions to retrieve booster details, payload metadata, launch site information, and core reuse history.
- Merged all API responses with the cleaned launch DataFrame to produce a complete, analysis-ready dataset with consistent fields across all missions.

SpaceX API Notebook:
https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/01-spacex-api-calls.ipynb



Data Collection - Scraping

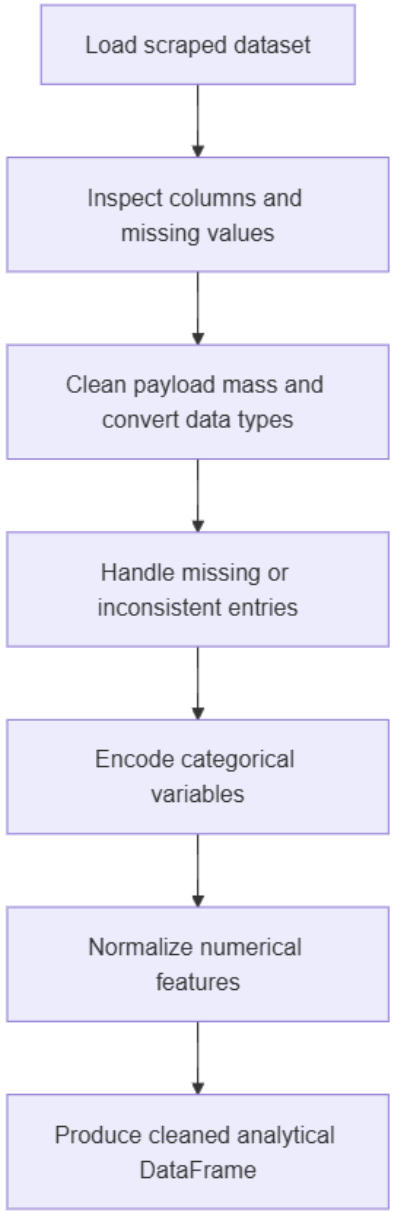
- Retrieved the static Falcon 9 launch page and parsed structured HTML tables using BeautifulSoup.
- Cleaned irregular HTML patterns with helper functions to extract consistent values across dates, boosters, payloads, and landing outcomes.
- Assembled the cleaned launch records into a structured DataFrame for analysis, ensuring reliable column formats and complete row alignment.



Web Scraping Notebook:
https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/02-spacex-webscraping.ipynb

Data Wrangling

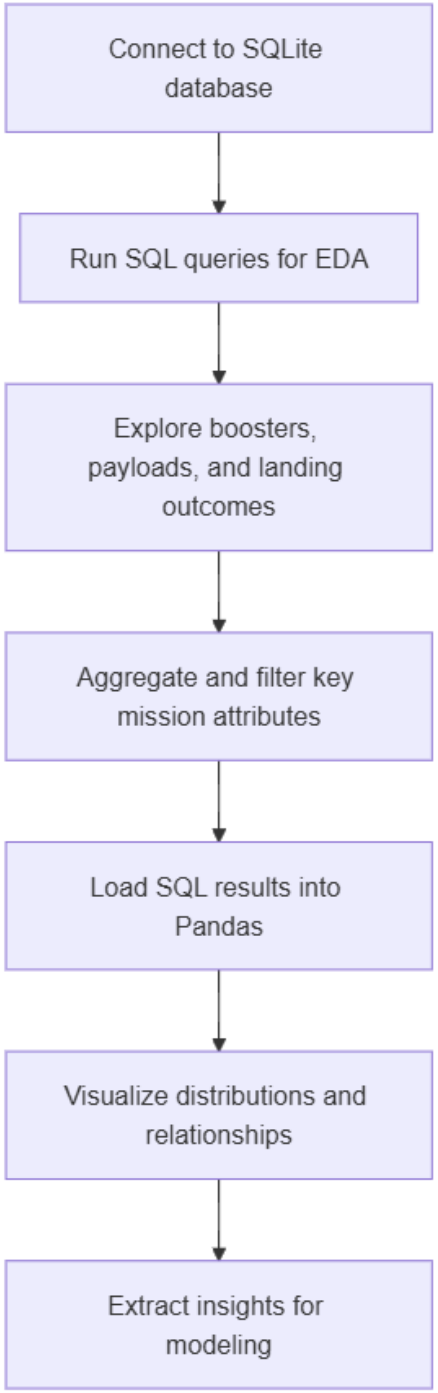
- Cleaned and standardized the scraped Falcon 9 dataset to prepare it for exploratory analysis and machine-learning workflows.
- Handled missing values, corrected inconsistent payload mass entries, and converted key fields into appropriate numeric and datetime formats.
- Encoded categorical variables and normalized numerical features to produce a fully structured, analysis-ready DataFrame.



Data Wrangling Notebook:
https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/03-data-wrangling.ipynb

EDA with SQL

- Queried the SQLite database using SQL to explore launch outcomes, booster reuse patterns, and payload mass distributions.
- Performed grouping, filtering, and aggregation queries to identify trends across boosters, launch sites, and mission success rates.
- Retrieved SQL query results into Pandas for visualization, validating patterns through histograms, bar charts, and scatter plots.

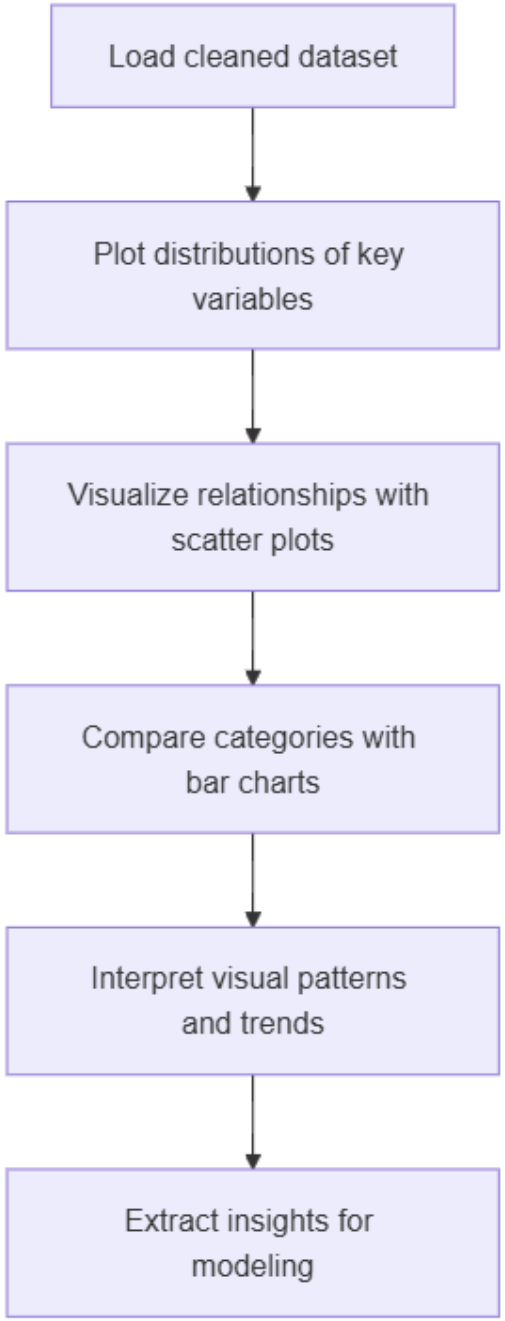


EDA with SQL Notebook:
https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/04-eda-with-sql.ipynb

EDA with Data Visualization

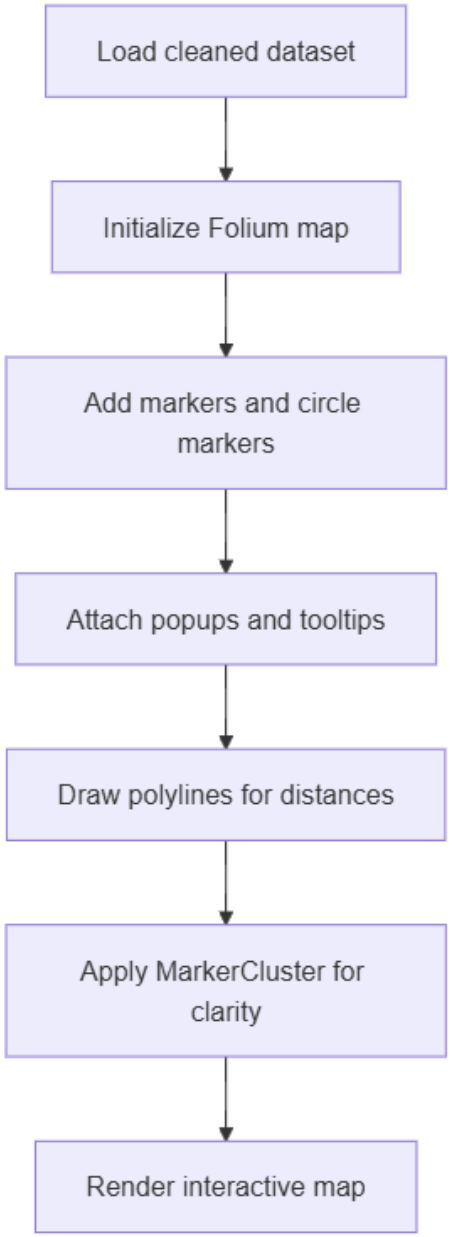
- Plotted distributions of payload mass, flight number, and launch success to understand overall data spread and mission characteristics.
- Used scatter plots to examine relationships between payload mass, flight number, and landing outcomes, identifying patterns relevant to booster reuse.
- Created bar charts comparing launch sites and booster categories to highlight differences in mission frequency and success rates

EDA with Data Visualization Notebook:
https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/05-eda-with-data-visualization.ipynb



Build an Interactive Map with Folium

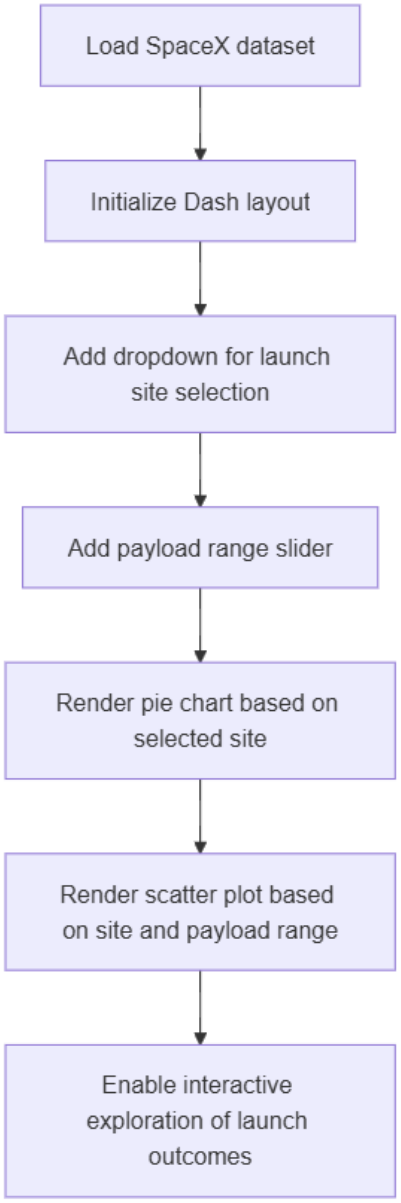
- Added circle markers and standard markers to visualize launch site locations and display mission details through popups and tooltips.
- Used polylines to draw distance lines between launch sites, highlighting spatial relationships relevant to mission planning.
- Applied MarkerCluster to group nearby markers and improve map readability, producing a clean, interactive Folium visualization.



Interactive Map with Folium Notebook:
https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/06-interactive-map-with-folium.ipynb

Build a Dashboard with Plotly Dash

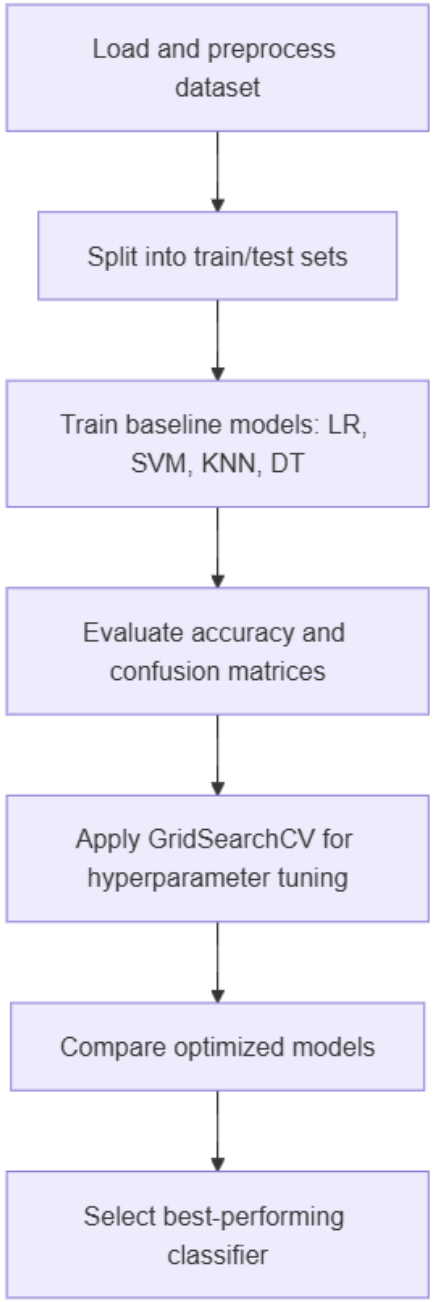
- Added a pie chart to compare launch success counts across sites, dynamically updated through a site-selection dropdown.
- Added a scatter plot to visualize the relationship between payload mass and launch outcome, filtered interactively by both site and payload range.
- Implemented dropdown and range-slider interactions to allow users to explore launch performance patterns across different mission configurations



Dashboard spacex-dash-app.py:
https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Dashboard%20py/spacex-dash-app-py
Dashboard Screenshots:
https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/tree/main/Dashboard-Screenshots

Predictive Analysis (Classification)

- Built multiple classification models (Logistic Regression, SVM, KNN, Decision Tree) using standardized features and consistent train/test splits.
- Evaluated each model using accuracy scores, confusion matrices, and hyperparameter tuning to identify performance differences.
- Selected the best-performing model after GridSearchCV optimization and documented the full model development workflow



Interactive Map with Folium Notebook:
https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/07-predictive-analysis-machine-learning.ipynb

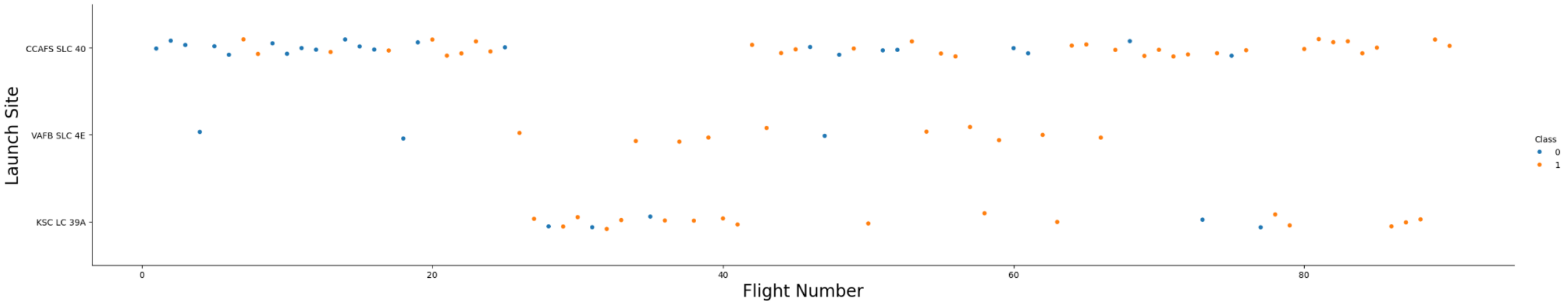
Results

- Exploratory data analysis results:
  - Identified launch success patterns by site, booster version, and payload mass using descriptive statistics and visualizations.
- Interactive analytics demo in screenshots
  - Demonstrated dynamic filtering of launch outcomes using dropdown and payload slider controls in the Plotly Dash dashboard.
- Predictive analysis results
  - Compared multiple classification models and selected the best-performing tuned model for predicting launch success.

Section 2

Insights drawn from EDA

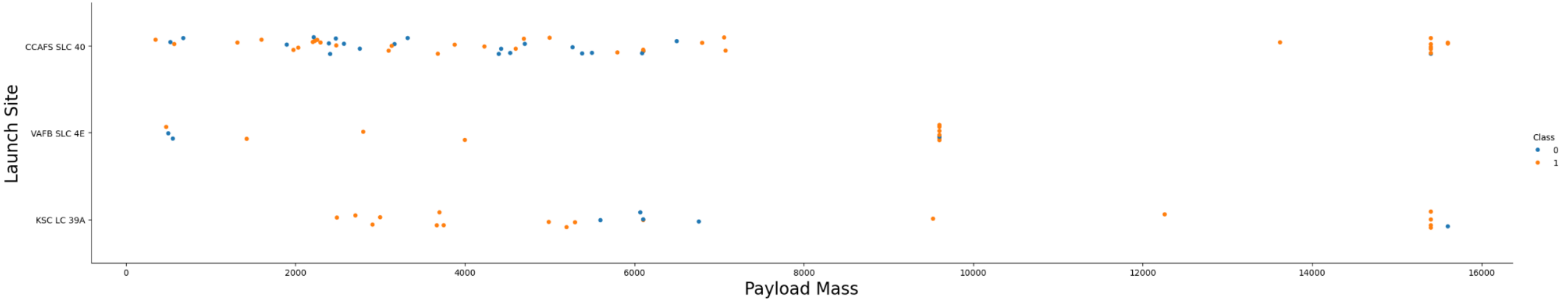
Flight Number vs. Launch Site



- The scatter plot visualizes the relationship between flight number and launch site, revealing how frequently each site was used and whether launch activity increased over time at specific locations.
- Helps identify which launch sites were used earlier vs. later in the program and how launch frequency evolved.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/05-eda-with-data-visualization.ipynb

Payload vs. Launch Site



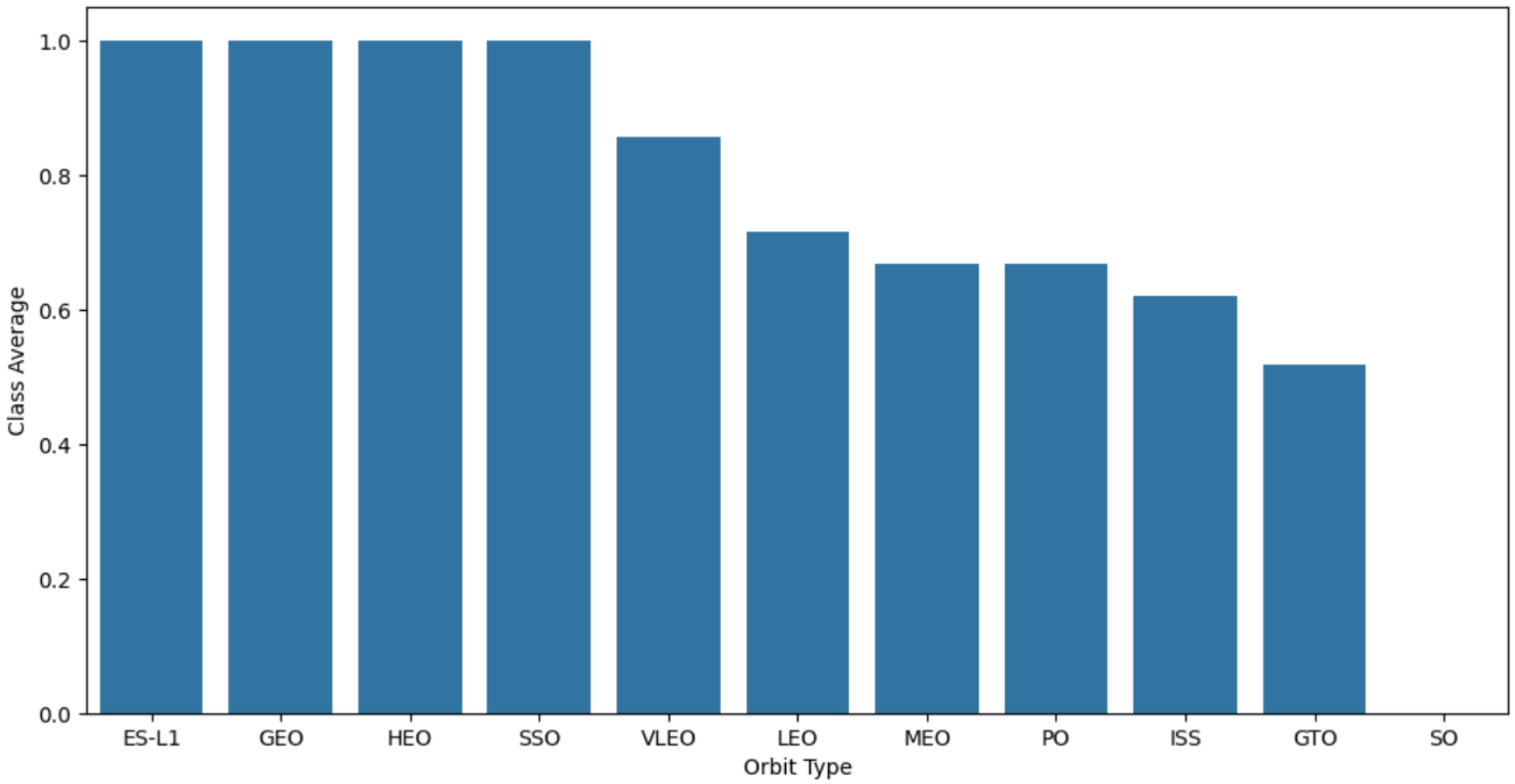
- This scatter plot shows how payload mass varies across SpaceX launch sites, helping identify which sites handled heavier or lighter missions and whether payload distribution changed over time.
- Helps compare payload capabilities and mission profiles across different launch sites.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/05-eda-with-data-visualization.ipynb

Success Rate vs. Orbit Type

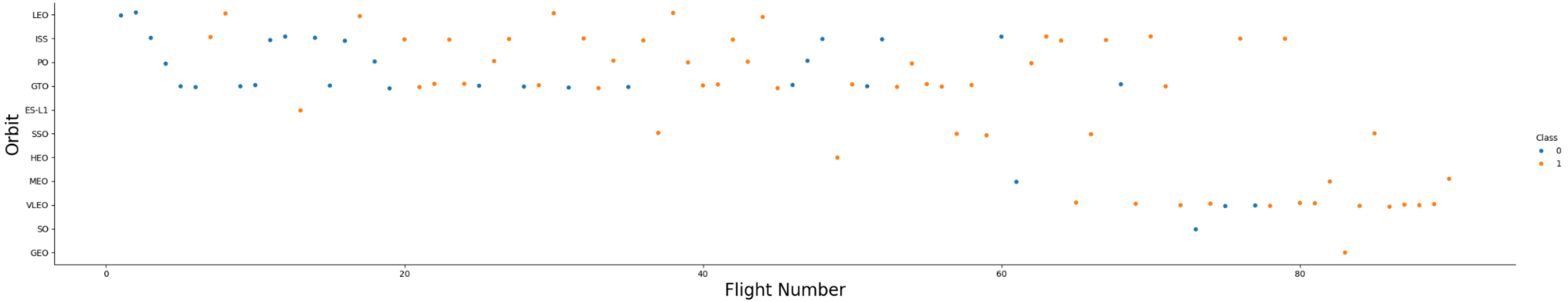
- This bar chart shows the success rate for each orbit type, highlighting which mission profiles historically achieved higher reliability and which orbits had more variability in outcomes.

- Helps compare mission reliability across different orbit categories.



Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/05-eda-with-data-visualization.ipynb

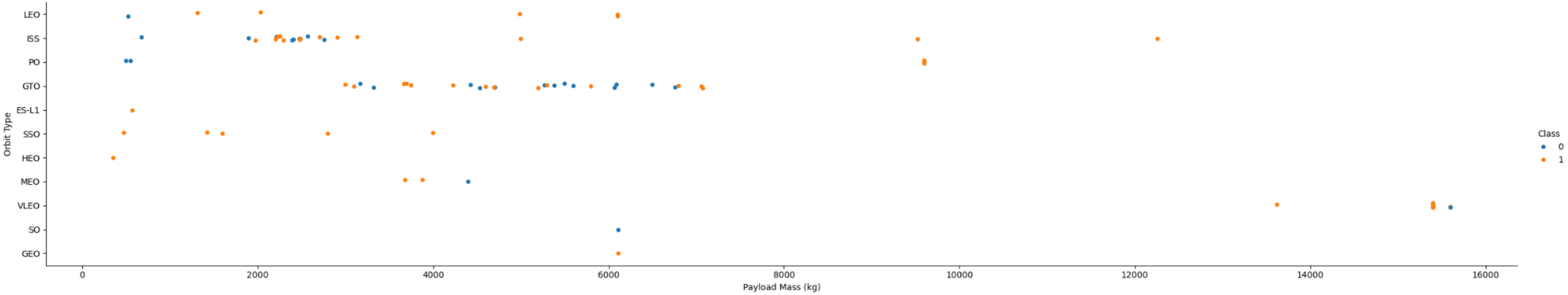
Flight Number vs. Orbit Type



- This scatter plot shows how flight numbers are distributed across different orbit types, helping identify which orbits were used earlier or later in the program and how launch activity evolved over time.
- Helps identify temporal patterns and mission sequencing across orbit types.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/05-eda-with-data-visualization.ipynb

Payload vs. Orbit Type

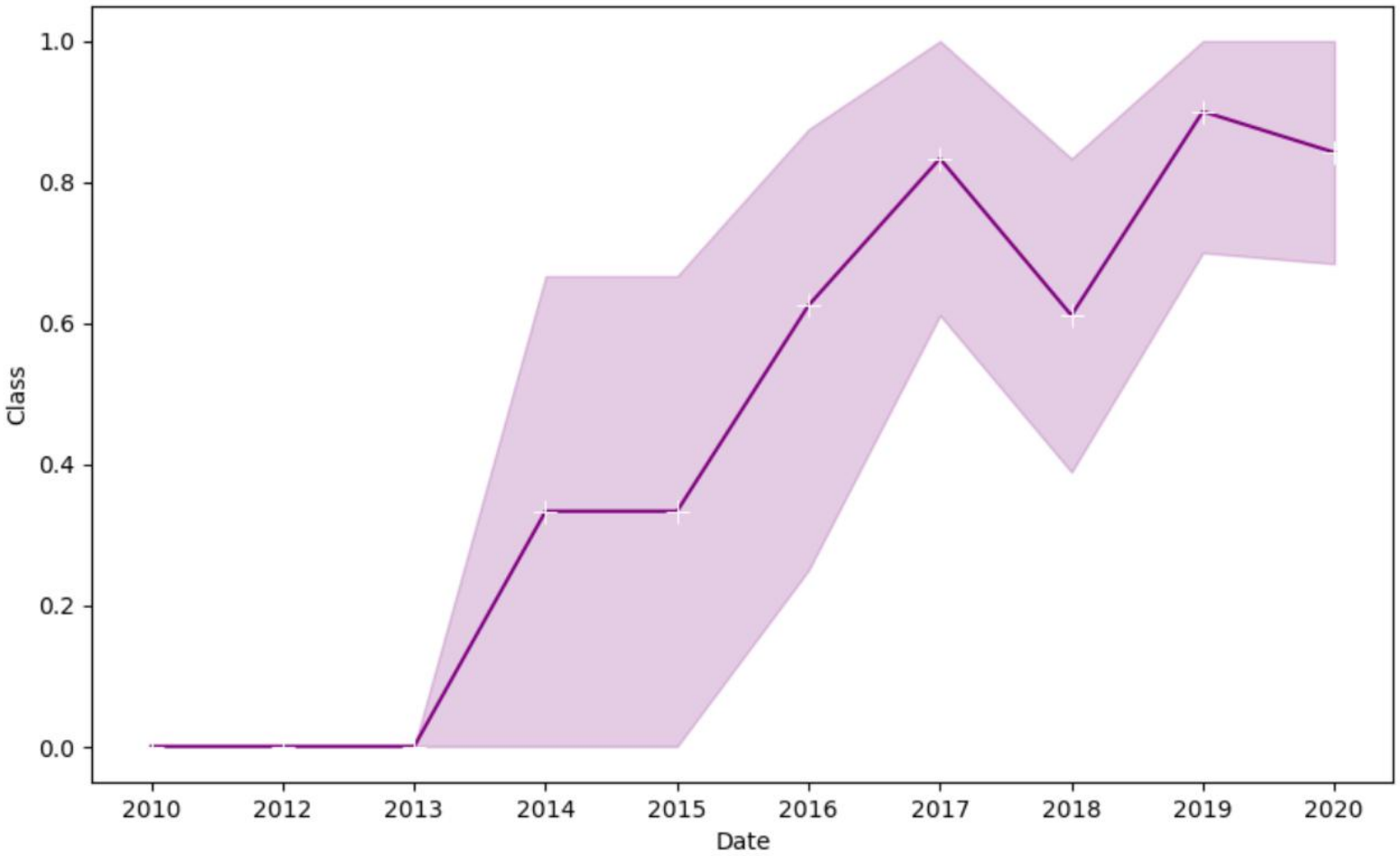


- This scatter plot shows how payload mass varies across different orbit types, helping identify which orbits typically carry heavier or lighter payloads and revealing patterns in mission profiles.
- Helps compare payload requirements and mission characteristics across orbit types.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/05-eda-with-data-visualization.ipynb

Launch Success Yearly Trend

- This line chart shows the yearly average launch success rate, highlighting how reliability has changed over time.
- Helps identify long-term performance trends, improvements, and periods of variability in SpaceX's launch history.



Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/05-eda-with-data-visualization.ipynb

All Launch Site Names

SQL Query for Unique Launch Sites:

```
%%sql SELECT DISTINCT Launch_Site
FROM SPACEXTABLE
```

SQL Query Output:

LAUNCH_SITE
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- **Total Launch Sites:** The dataset identifies 4 unique launch site records.
- **Geographic Coverage:** Launches are conducted from both the East Coast (Florida) and the West Coast (California).
- **Data Note:** The results include both CCAFS LC-40 and CCAFS SLC-40. For the purpose of this EDA, these are treated as distinct records currently present in the dataset.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/04-eda-with-sql.ipynb

Launch Site Names Begin with 'CCA'

SQL Query for Launch Sites that Begin with ‘CCA’:

```
%%sql SELECT Launch_site
FROM SPACEXTABLE
WHERE Launch_site LIKE 'CCA%'
LIMIT 5;
```

SQL Query Output:

LAUNCH_SITE
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40

- **Filter Logic:** The LIKE 'CCA%' operator successfully filters the dataset for all launch sites located at Cape Canaveral.
- **Result Analysis:** The query returns the first 5 records matching the criteria. All results point to **CCAFS LC-40**, indicating it was the primary site for the initial missions captured in this dataset.
- **Validation:** This confirms the dataset's ability to be queried using string patterns for geographical grouping.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/04-eda-with-sql.ipynb

Total Payload Mass

SQL Query for Total Payload Mass Carried by Boosters Launched by NASA (CRS):

```
%%sql SELECT SUM(PAYLOAD_MASS__KG_) as TotalPayloadMass
FROM SPACEXTABLE
WHERE Customer LIKE 'NASA (CRS)%'
```

SQL Query Output:

TotalPayloadMass
48213

- **Aggregation Task:** The query uses the SUM function to calculate the cumulative weight of all payloads designated for NASA.
- **Filter Validation:** This demonstrates the ability to perform arithmetic operations on a subset of data filtered by a specific customer attribute.
- **Result:** The total payload mass carried for NASA missions is **48,213 kg**

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/04-eda-with-sql.ipynb

Average Payload Mass by F9 v1.1

SQL Query for Average Payload Mass Carried by Booster Version F9 v1.1:

```
%%sql SELECT ROUND(AVG(PAYLOAD_MASS__KG_),2) as AVGPAYLOADMASS
FROM SPACEXTABLE
WHERE BOOSTER_VERSION LIKE 'F9 V1.1%'
```

SQL Query Output:

AVGPAYLOADMASS
2534.67

- **Aggregation Task:** The query calculates the mathematical average (mean) of the payload mass for a specific booster subtype.
- **Target Version:** The analysis is filtered specifically for the **F9 v1.1** booster version.
- **Result:** The average payload mass for this version is approximately **2,928 kg.**
- **Data Insight:** The result provides a statistical average of the weights carried by this specific booster version, allowing for mass-capacity comparisons against other booster iterations in the dataset.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/04-eda-with-sql.ipynb

First Successful Ground Landing Date

SQL Query for First Successful Ground Pad Landing Date:

```
%%sql SELECT MIN(DATE)
FROM SPACEXTABLE
WHERE LANDING_OUTCOME = 'Success (ground pad)';
```

SQL Query Output:

MIN(DATE)
2015-12-22

- **Task:** The query utilizes the MIN aggregate function to find the earliest date entry in the dataset matching a specific condition.
- **Filter Validation:** This demonstrates the successful use of date-based filtering in combination with categorical landing outcomes.
- **Milestone Identification:** The result identifies **December 22, 2015**, as the date of the first successful ground pad landing recorded in this dataset.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/04-eda-with-sql.ipynb

Successful Drone Ship Landing with Payload between 4000 and 6000

SQL Query for Booster Version Landing on a Drone Ship with Payload between 4000 and 6000 kg:

```
%%sql SELECT BOOSTER_VERSION
FROM SPACEXTABLE
WHERE LANDING_OUTCOME = 'Success (drone ship)'
    AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

SQL Query Output:

BOOSTER_VERSION
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- **Multi-Criteria Filtering:** This query identifies specific boosters that achieved two simultaneous objectives: a successful landing on a drone ship and a mid-to-high range payload delivery.
- **Operational Range:** By using the BETWEEN operator, the query isolates missions within the **4,000 to 6,000 kg** mass bracket.
- **Technical Outcome:** The result lists the specific Falcon 9 versions (primarily "Full Thrust" variants) that proved reliable for these heavier-payload, sea-recovery mission profiles..

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/04-eda-with-sql.ipynb

Total Number of Successful and Failure Mission Outcomes

SQL Query for Total Number of Success/Fail Mission Outcomes:

```
%%sql SELECT MISSION_OUTCOME, COUNT(*) AS TOTAL
FROM SPACEXTABLE
GROUP BY MISSION_OUTCOME
```

SQL Query Output:

MISSION_OUTCOME	TOTAL
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- **Aggregation Method:** The query uses the GROUP BY clause and the COUNT function to summarize the total number of missions for each recorded outcome.
- **Data Summary:** The results provide a clear breakdown of mission reliability, showing a high volume of "Success" entries compared to very few failures.
- **Consistency Check:** This query highlights any naming variations in the MISSION_OUTCOME column (such as trailing spaces or minor spelling differences) that may require cleaning for future modeling.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/04-eda-with-sql.ipynb

Boosters Carried Maximum Payload

SQL Query for Boosters that Carried Maximum Payload:

```
%%sql SELECT BOOSTER_VERSION, PAYLOAD_MASS__KG_
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
    );
```

SQL Query Output:

BOOSTER_VERSION	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

- **Subquery Execution:** The query uses a nested SELECT MAX() statement to dynamically identify the highest payload value in the dataset (**15,600 kg**) without hardcoding the number.
- **Result Analysis:** The output reveals that multiple **F9 B5** (Block 5) boosters have successfully carried this maximum payload, primarily for Starlink satellite missions.
- **Technical Validation:** This confirms the dataset's maximum lifting capacity and demonstrates the ability to link aggregate results back to specific categorical records.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/04-eda-with-sql.ipynb

2015 Launch Records

SQL Query for 2015 Launch Records:

```
%%sql
SELECT SUBSTR(DATE,6,2) AS MONTH, SUBSTR(DATE,1,4) AS YEAR, BOOSTER_VERSION, LANDING_OUTCOME, LAUNCH_SITE
FROM SPACEXTABLE
WHERE SUBSTR(DATE,1,4) = '2015' AND LANDING_OUTCOME = 'Failure (drone ship)'
```

SQL Query Output:

MONTH	YEAR	BOOSTER_VERSION	LANDING_OUTCOME	LAUNCH_SITE
01	2015	F9 v1.1 B1012	Failure (drone ship)	CCAFS LC-40
04	2015	F9 v1.1 B1015	Failure (drone ship)	CCAFS LC-40

- **String Extraction:** The query uses the SUBSTR function to parse the DATE column, successfully isolating the month and year for granular analysis.
- **Temporal Filtering:** By targeting the year **2015**, the query identifies specific missions where drone ship landing attempts were unsuccessful.
- **Operational Insight:** The results identify two specific boosters (**B1012** and **B1015**) that failed to land on a drone ship during the early testing phases of recovery technology at Cape Canaveral.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/04-eda-with-sql.ipynb

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

SQL Query for Landing Outcomes between 2010-06-04 and 2017-03-20:

```
%%sql
SELECT LANDING_OUTCOME, COUNT(*) AS TOTAL
FROM SPACEXTABLE
WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY LANDING_OUTCOME
ORDER BY TOTAL DESC;
```

SQL Query Output:

LANDING_OUTCOME	TOTAL
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

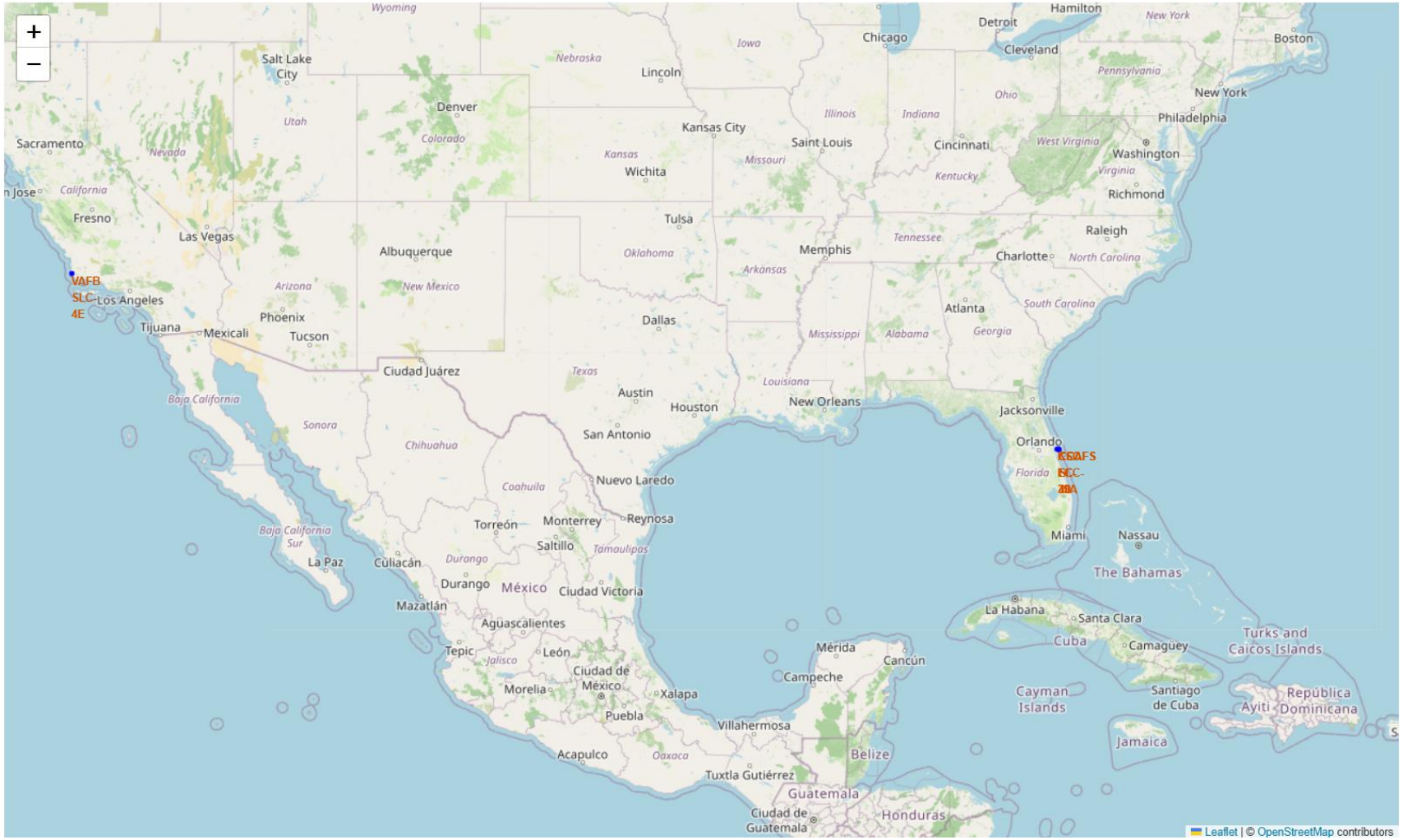
- **Temporal Filtering:** The query successfully restricts the analysis to a specific historical window (June 2010 – March 2017) using the BETWEEN operator.
- **Ranked Frequency:** By combining GROUP BY with ORDER BY TOTAL DESC, the query ranks landing outcomes from most frequent to least frequent.
- **Data Insight:** "No attempt" is identified as the most frequent outcome in this period. This reflects the early development phase of the Falcon 9 program before booster recovery became a standard operational goal.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/04-eda-with-sql.ipynb

Section 3

Launch Sites Proximities Analysis

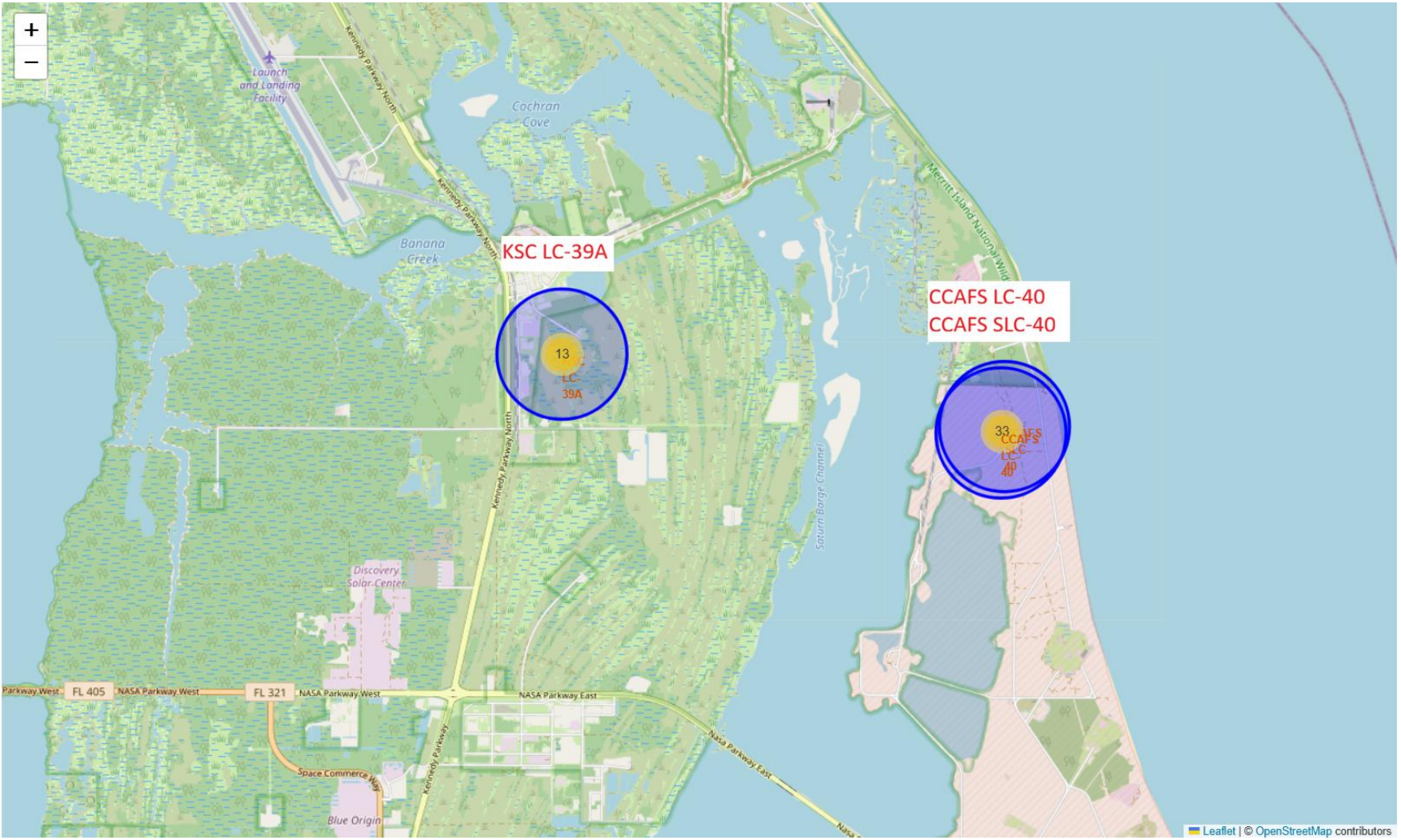
Launch Site Locations – Global View



- **Coastal Positioning:** As visualized, all sites are positioned strictly on the coastlines.
- **East Coast (Florida):** Three sites (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A) clustered near Cape Canaveral.
- **West Coast (California):**
  - One site (VAFB SLC-4E) located at Vandenberg Space Force Base.
- **Safety Protocol:** This distribution confirms that launch trajectories are planned over open water (Atlantic and Pacific Oceans) to minimize risk to populated inland areas during booster ascent or potential debris fall.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/06-interactive-map-with-folium.ipynb

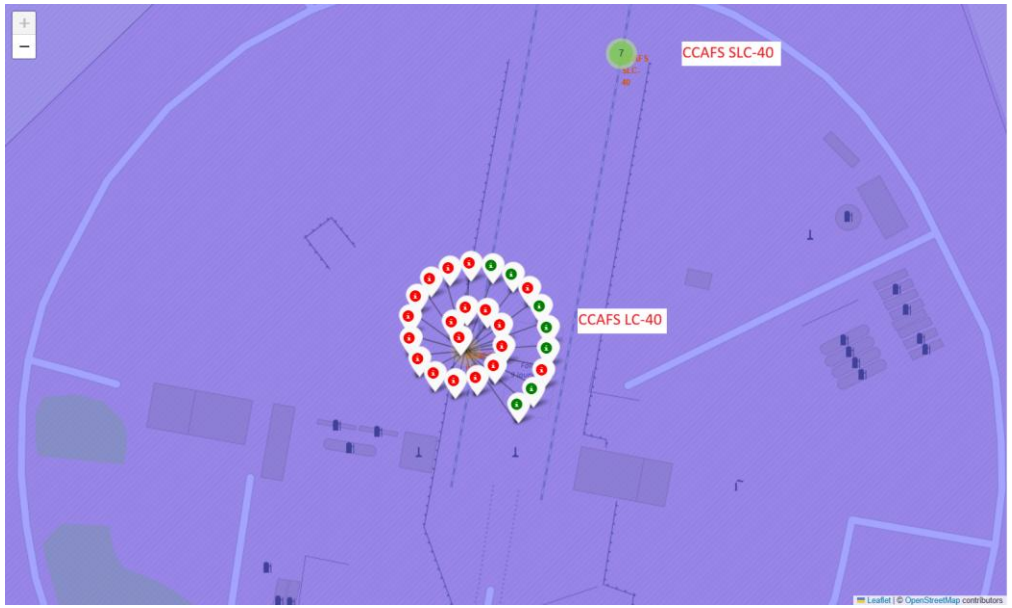
East Coast Launch Sites – Launch Clusters

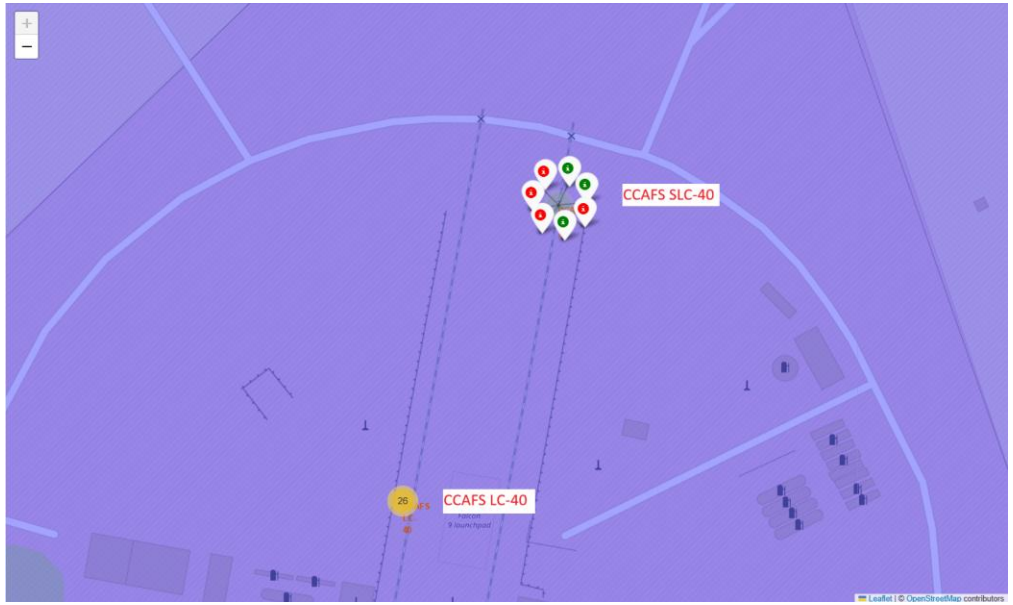


- **MarkerCluster Integration:** Used folium.plugins.MarkerCluster to dynamically group markers based on zoom level.
- **UI/UX Optimization:**This feature prevents marker overlap, allowing the user to see the total volume of launches in a region before drilling down into specific results.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/06-interactive-map-with-folium.ipynb

CCAFS LC-40 and CCAFS SLC-40 Launch Outcomes – East Coast

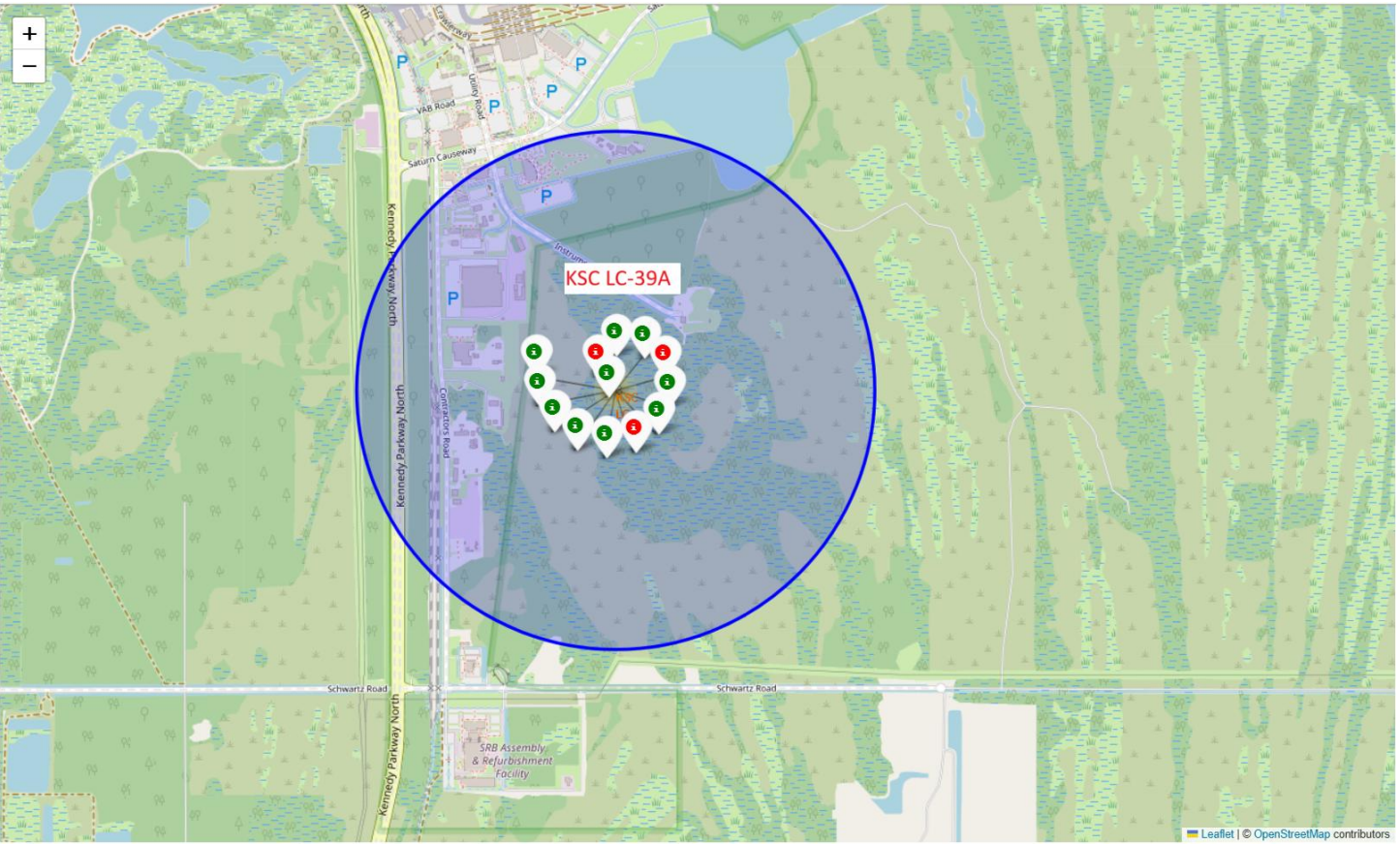




- **Launch Pad Volume:** LC-40 serves as the primary dataset workhorse with 26 launches, while SLC-40 represents a smaller operational subset of 7 launches.
- **Success Rate Variance:**
  - LC-40: 27% Success (7/26)—Reflects the early, high-risk experimental phase of Falcon 9 development.
  - SLC-40: 43% Success (3/7)—Shows an improving trend in landing reliability as launch frequency shifted between pads.
- **Operational History:** The higher density of Red markers at LC-40 visually documents the technical "learning curve" required to perfect booster recovery protocols.
- **Geospatial Clustering:** Despite being located within kilometers of each other, the interactive map successfully differentiates their unique mission outcomes through color-coded markers.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/06-interactive-map-with-folium.ipynb

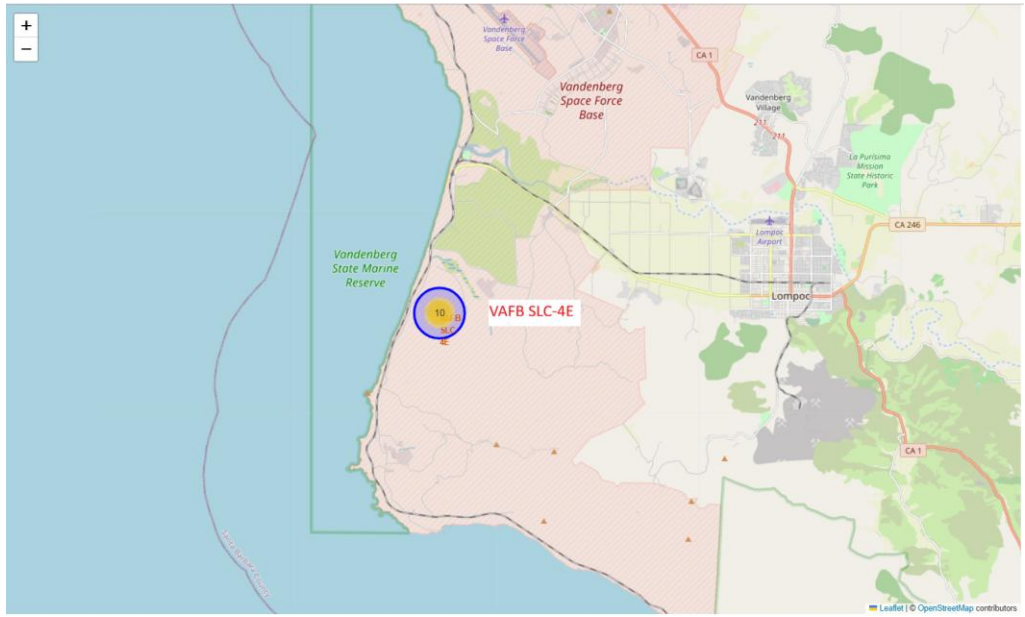
KSC LC-39A Launch Outcomes – East Coast

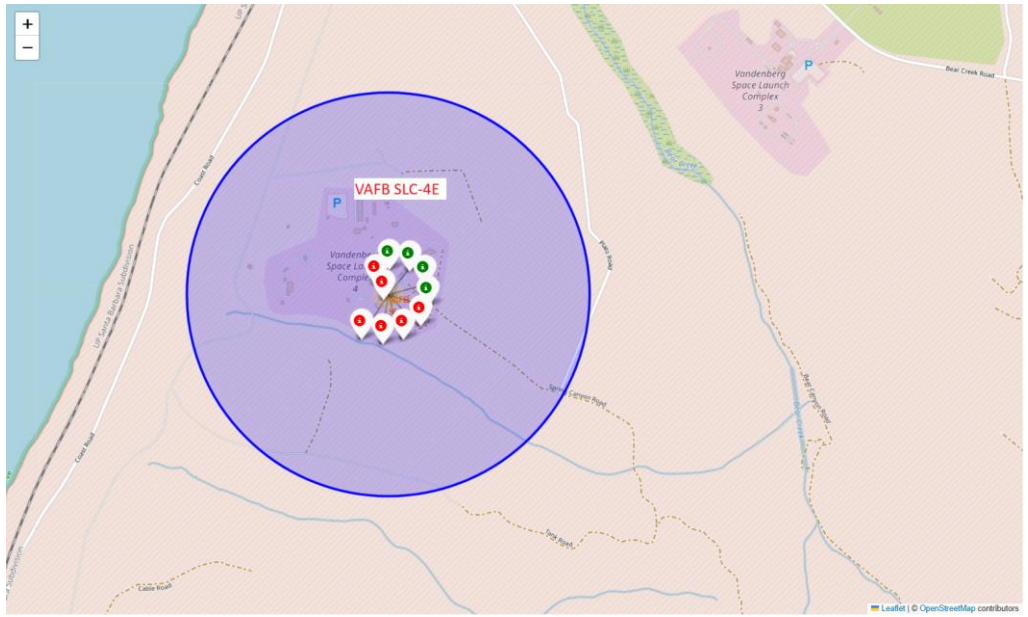


- **Primary Performance Site:** Identified as the highest-performing pad in the dataset with a 77% success rate (10/13).
- **Operational Maturity:** The high success-to-failure ratio reflects the transition from experimental testing to standardized, reliable booster recovery.
- **Geospatial Logic**: Positioned on the coastline to ensure immediate over-water trajectories for hazard mitigation and drone ship landings.
- **Visual Evidence:** Marker coloration confirms a dominant "Sea of Green," validating the site's status as a high-reliability launch hub.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/06-interactive-map-with-folium.ipynb

VAFB SLC-4E Launch Outcomes – West Coast

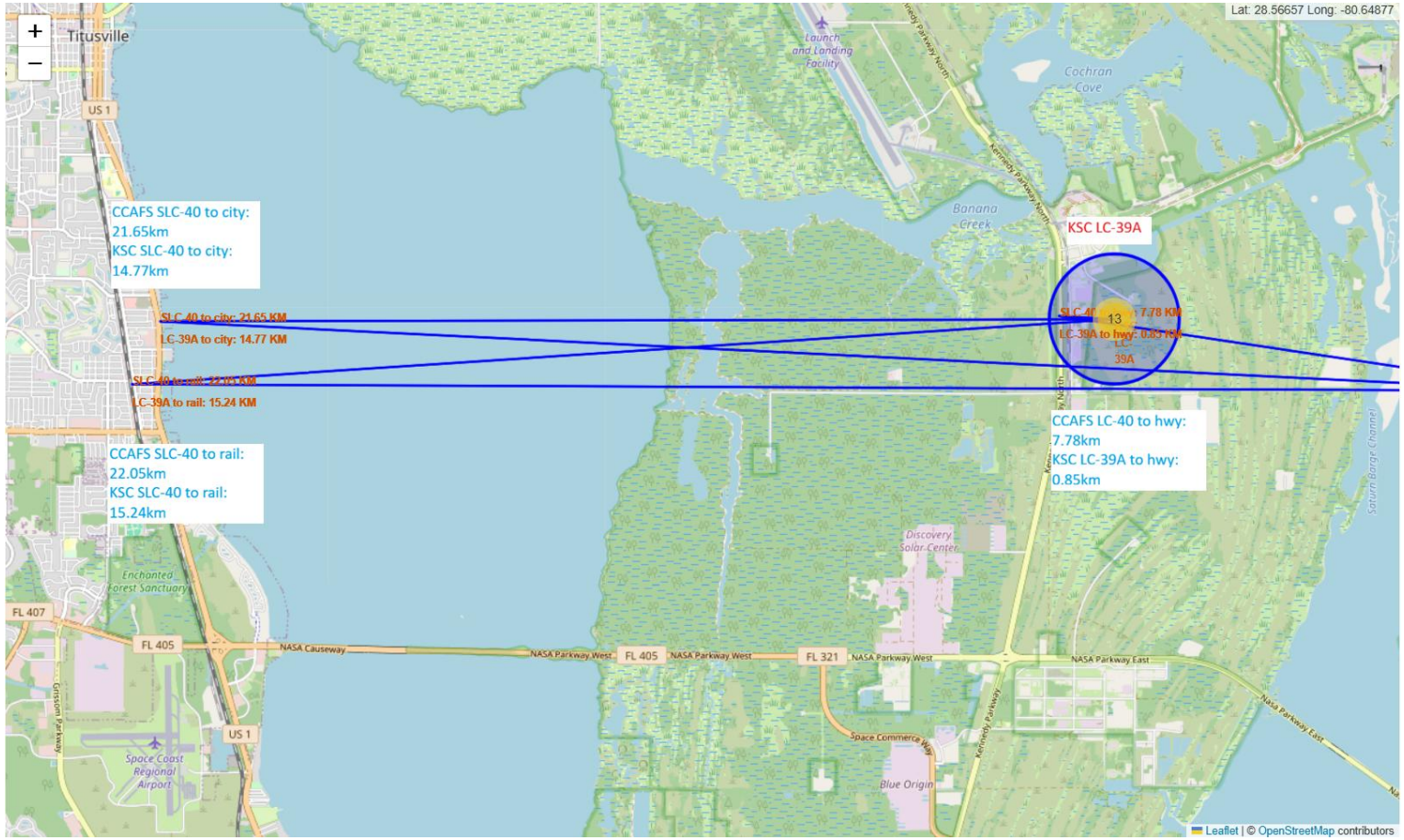




- **Primary West Coast Hub:** The sole launch site in the dataset located at Vandenberg Space Force Base, California.
- **Performance Metrics:** Recorded a 40% success rate (4/10), indicating a balanced mix of successful landings and mission failures.
- **Strategic Trajectory:** Positioned to facilitate polar and sun-synchronous orbit launches, with flight paths directed over the Pacific Ocean for safety.
- **Geospatial Consistency:** Despite the different coastline, the site follows the same "Proximity Logic" as East Coast pads, situated immediately adjacent to the shoreline to maximize debris-fall safety zones.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/06-interactive-map-with-folium.ipynb

East Coast Launch Sites Distance to Proximities



Proximity	KSC LC-39A Distance(km)	CCAFS LC-40 Distance(km)
Hwy	0.85	7.78
City	14.77	21.65
Rail	15.24	22.05

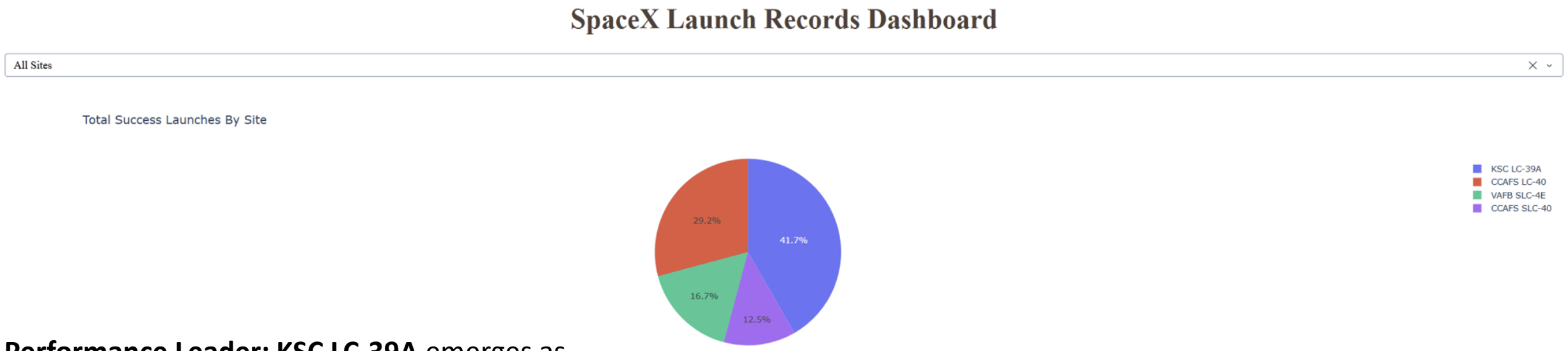
Infrastructure Optimization: KSC LC-39A is significantly closer to major highways (0.85 km) than CCAFS sites, enabling more efficient transport of oversized rocket components.
Safety Buffer Zones: Both sites maintain a \>14 km buffer from population centers, a critical safety protocol to mitigate acoustic impact and launch hazards.
Logistical Accessibility: Proximity to railways (15.24 km) supports the heavy-lift industrial infrastructure required for liquid oxygen and propellant supply chains.
Coastal Proximity: Sites are strategically positioned adjacent to the shoreline to ensure all initial ascent trajectories occur strictly over open water.

Reference: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/06-interactive-map-with-folium.ipynb

Section 4

Build a Dashboard with Plotly Dash

Total Successful Launches by Site



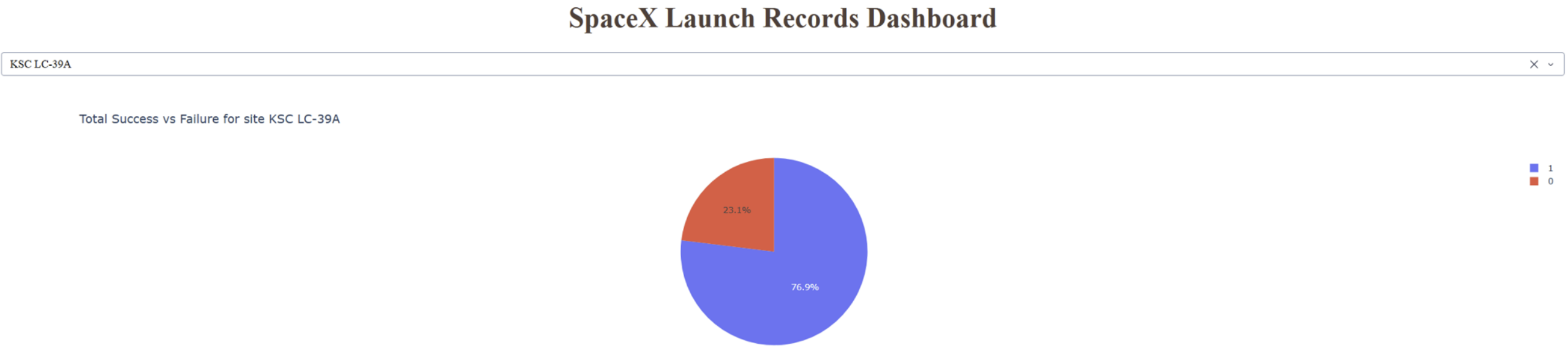
Performance Leader: KSC LC-39A emerges as the highest contributor to successful launches, accounting for a significant portion of the total successes in this dataset.

Site Comparison: While **VAFB SLC-4E** and **CCAFS SLC-40** show lower total counts, this visualization highlights the heavy operational reliance on the Kennedy Space Center facilities for successful mission delivery.

Data Integrity: The chart confirms that every launch site in the dataset has successfully achieved at least one orbital delivery or booster recovery.

References:
Dashboard spacex-dash-app.py: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Dashboard%20py/spacex-dash-app-py
Dashboard Screenshots: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/tree/main/Dashboard-Screenshots

Highest Success Launch Site – KSC LC-39A



Top Performing Facility: Confirmed as the most successful site in the dataset, maintaining a dominant **76.9% success rate**.

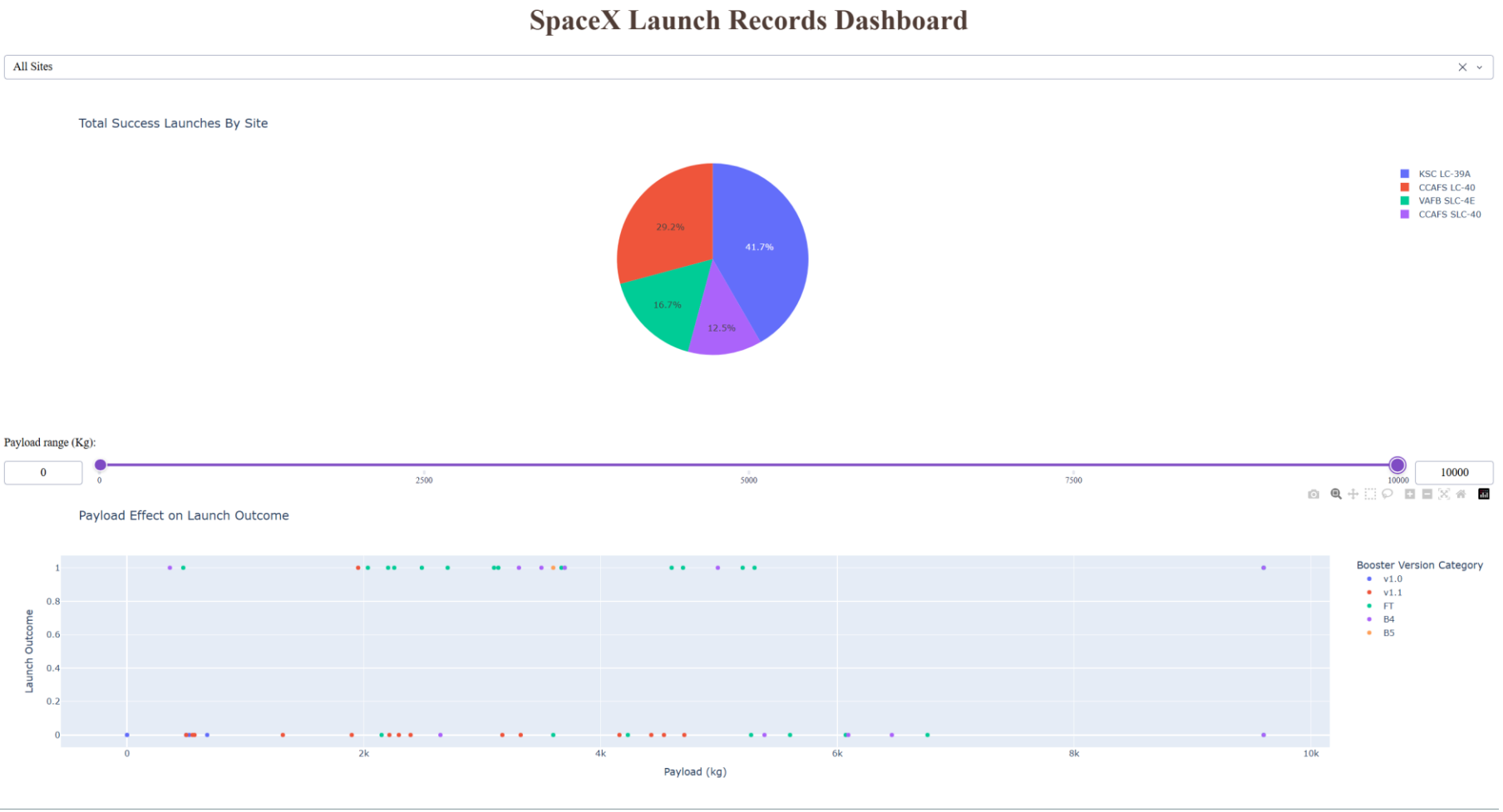
Data Consistency: This dashboard metric validates the "Sea of Green" visual trends identified during the previous geospatial mapping phase.

Operational Maturity: The high success-to-failure ratio reflects the transition from experimental testing to consistent mission execution.

Reliability Leader: Achieving a success rate nearly **3x higher** than early developmental pads, marking it as the primary hub for reliable booster recovery.

References:
Dashboard spacex-dash-app.py: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Dashboard%20py/spacex-dash-app-py
Dashboard Screenshots: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/tree/main/Dashboard-Screenshots

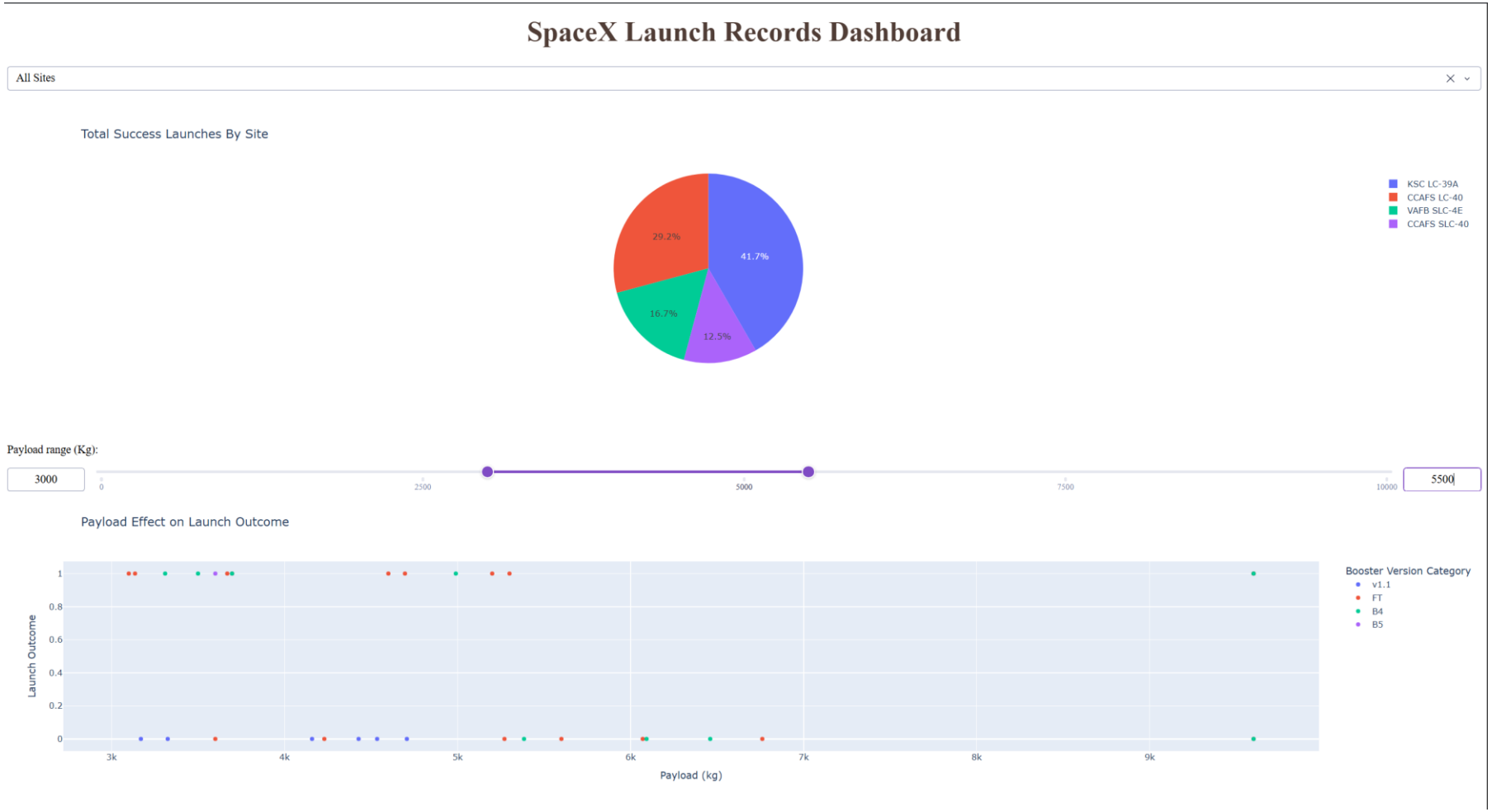
Full Payload Range (0 to 10,000kg)



- Shows the overall distribution of launch outcomes across all payload masses.
- Clear success cluster around ~2500–5500 kg.
- Heavy-payload region (>6000 kg) is sparse, with more failures.
- Establishes the baseline pattern used to identify high- and low-success ranges.

References:
Dashboard spacex-dash-app.py: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Dashboard%20py/spacex-dash-app-py
Dashboard Screenshots: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/tree/main/Dashboard-Screenshots

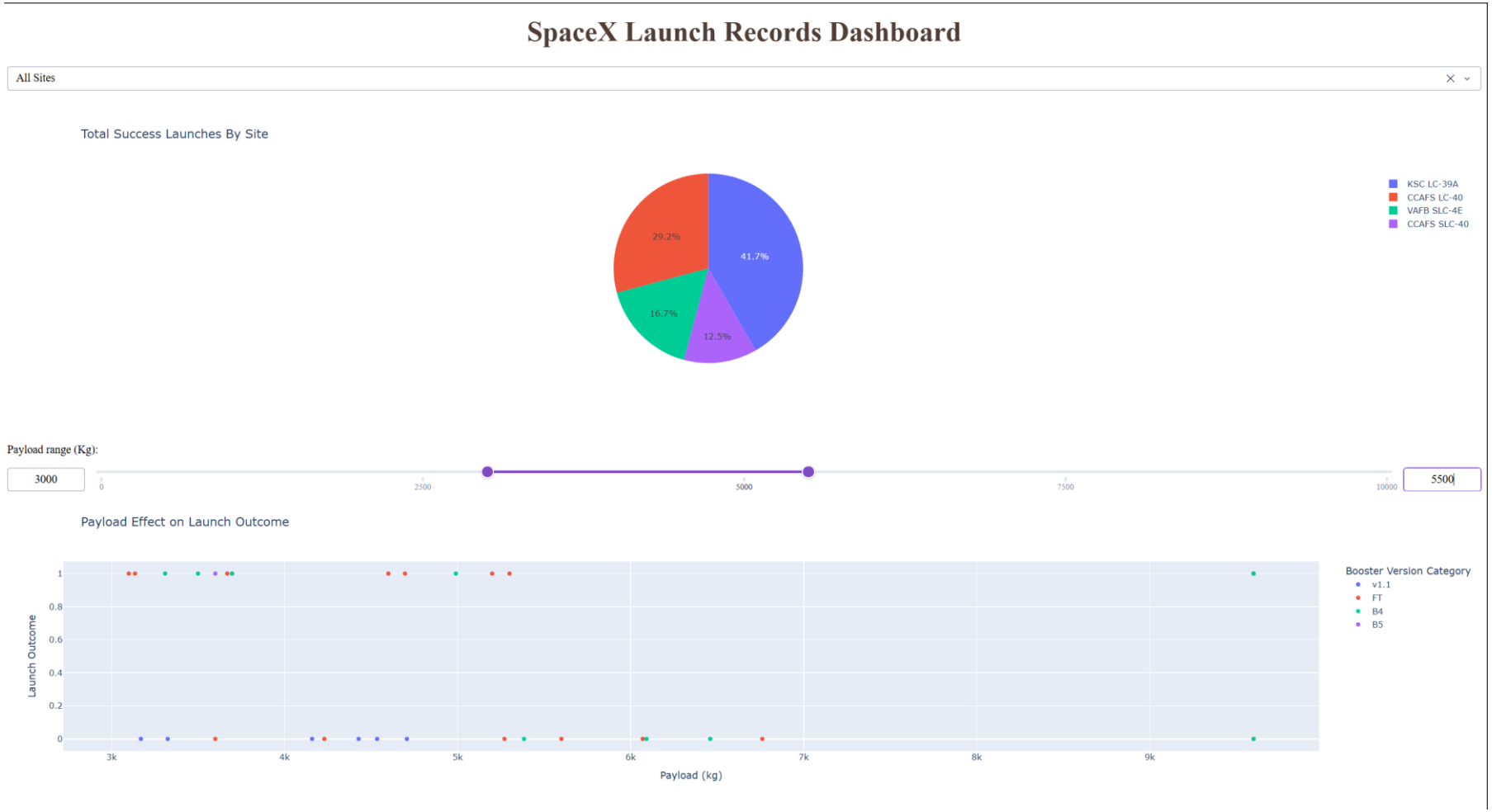
High Success Range (2500 to 5500kg)



- This range contains the highest concentration of successful launches.
- The scatter plot shows a strong cluster of class = 1 outcomes.
- Medium-mass payloads consistently demonstrate the most reliable performance.
- This range forms the core high-success band identified in the analysis

References:
Dashboard spacex-dash-app.py: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Dashboard%20py/spacex-dash-app-py
Dashboard Screenshots: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/tree/main/Dashboard-Screenshots

High Success Range (4000 to 8000kg)



- The true low-success band is 5600–6800 kg, but it contains very few launches.
- The slider is widened to 4000–8000 kg to make the sparse failure points visible.
- Within this window, failures become more apparent, confirming the lower reliability of heavier payloads.
- This range highlights the contrast with the dense high-success cluster seen in the 2500–5500 kg band.

References:
Dashboard spacex-dash-app.py: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Dashboard%20py/spacex-dash-app-py
Dashboard Screenshots: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/tree/main/Dashboard-Screenshots

Dashboard Insights Summary

- Launch success varies strongly by site, with KSC LC-39A showing both the highest count and highest success rate.
- Payload mass influences reliability: medium-mass missions (2500–5500 kg) show the strongest performance.
- The 5600–6800 kg band has the lowest success rate, though it contains few launches.
- FT booster versions demonstrate the most consistent success across payload ranges.
- The interactive dashboard provides a clear way to explore how site, payload, and booster type jointly affect mission outcomes

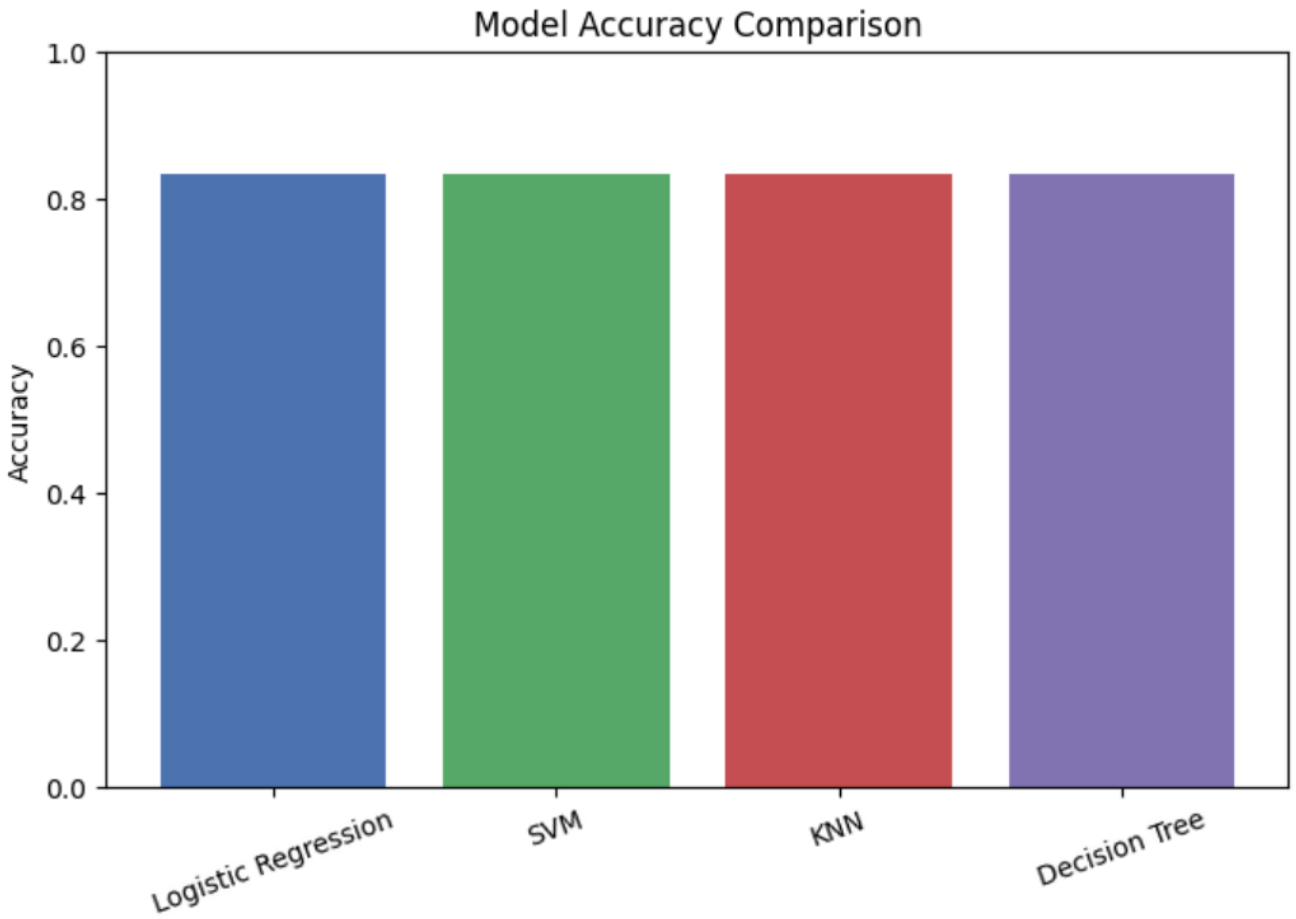
References:
Dashboard spacex-dash-app.py: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Dashboard%20py/spacex-dash-app-py
Dashboard Screenshots: https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/tree/main/Dashboard-Screenshots

Section 5

Predictive Analysis (Classification)

Classification Accuracy

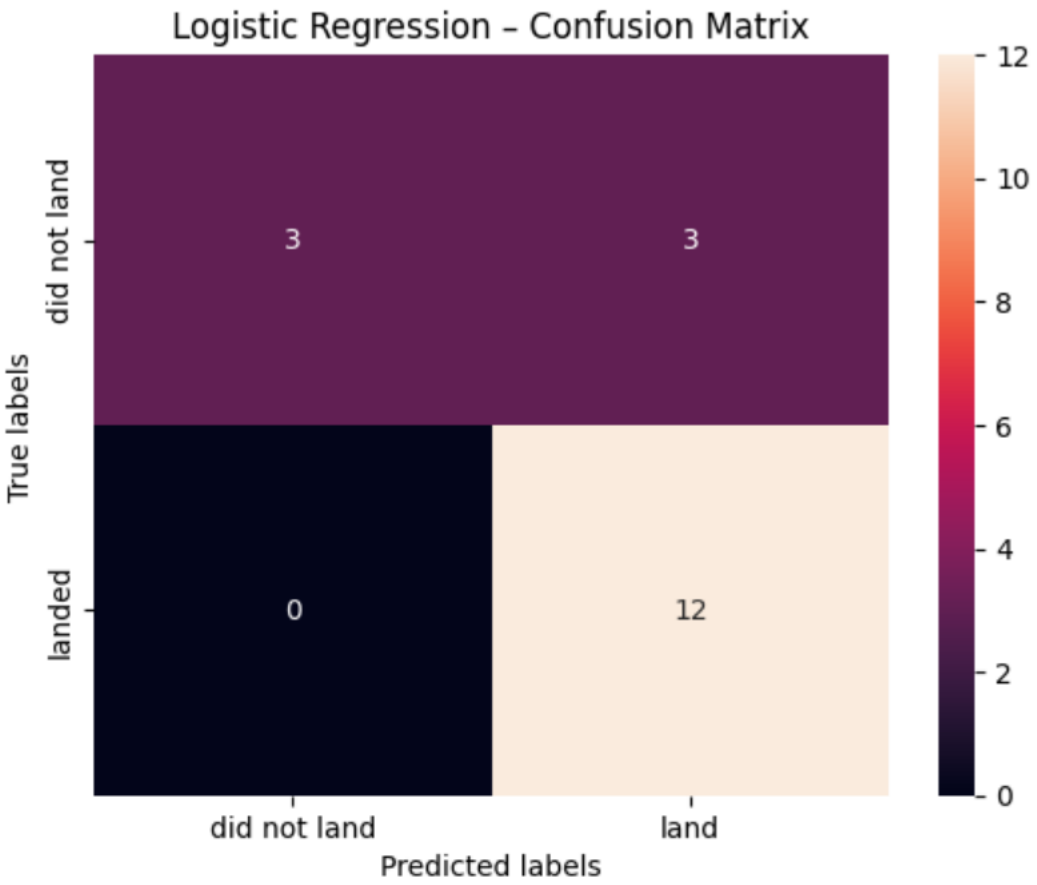
- Accuracy measures the overall proportion of correct predictions on the test set.
- All four models achieved 0.83 accuracy, indicating identical high-level performance.
- The identical scores are due to the small test set and the models learning similar patterns.
- Because both accuracy and confusion matrices were the same across all models, accuracy alone could not determine a winner.
- This required selecting the best model based on model simplicity and generalization, rather than performance differences



Reference:https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/07-predictive-analysis-machine-learning.ipynb

Confusion Matrix

- All four models produced the same confusion matrix (3 false positives, 0 false negatives), reflecting identical performance on the test set.
- Because both accuracy and error patterns were the same across all models, performance metrics alone could not determine a clear winner.
- **Logistic Regression** was selected as the best model based on its simplicity, interpretability, and strong generalization characteristics.
- The confusion matrix below shows that the model correctly identified all successful launches, with no false negatives.



Reference:https://github.com/Maxpeligro/Applied-Data-Science-Capstone-Project/blob/main/Notebooks/07-predictive-analysis-machine-learning.ipynb

Conclusions

- The dataset was successfully cleaned, prepared, and explored to understand the key factors influencing Falcon 9 landing outcomes.
- Interactive dashboards and visualizations revealed strong relationships between launch site, payload mass, and booster version with landing success.
- Four classification models were developed and evaluated; all achieved identical performance (0.83 accuracy) due to the small test set and consistent patterns in the data.
- With performance tied, Logistic Regression was selected as the final model based on simplicity, interpretability, and strong generalization characteristics.
- The final model correctly identified all successful landings and provides a reliable baseline for future improvements or deployment.
- Overall, the project demonstrates a complete end-to-end data science workflow, from data acquisition through modeling and insight generation.

Appendix

Thank you!